# Deposit funding and cross-border sovereign portfolios

Diagnosis, tractable alternatives, and a stress test for the global model

Working note, branch `analysis` (from `main` at `55e174d`)

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## Summary

**Issue 1 (deposit funding rate).** In every deposit closure the code offers, the deposit rate is an *equilibrium price that falls* when sovereign risk rises, and that fall cancels 51–61% of the credit-spread increase before it reaches firms' working-capital cost. The model's own output decomposition attributes $-0.179\%$ of impact output to the credit spread and $+0.104\%$ to the deposit rate, for a total of $-0.073\%$. The credit-spread channel alone is close to Bocola's open-economy benchmark of $-0.157\%$; the funding-rate offset is what separates the model from it. The two closures also cost tractability in different ways. The national closure needs three solver devices and cannot host the TPI. The union closure needs an extra state whose stationarity rests on a borrowed premium. And the choice between the two flips the sign of the 25-quarter cumulative output response ($-3.9$ against $+0.5$). In the stress test, the union closure's uncalibrated premium $\kappa_{\text{nfa}}$ moves impact output four times as much per log unit as the portfolio cost $\psi$.

**Issue 2 (cross-border holdings).** The adjustment cost $\psi$ does four jobs at once: it selects a portfolio that single-$\lambda$ banks leave indeterminate; it sets the return semi-elasticity of cross-border holdings, which is exactly $1/\psi$; under national funding it is the inverse elasticity of the country's *entire* external adjustment; and it keeps the default branch solvable, since below $\psi \approx 1$ the union economy has no solution on the current ladder. Over the solvable range ($\psi \in [1, 32]$, quick grid) impact output moves by a third of its value, while the deposit-rate channel is about five times larger. Two calibration faults compound this. The anchor is a *level*, so a growing debt stock is absorbed at home by construction. And the symmetric gross position gives the D bank a foreign share of its sovereign book of 20%, against 0.75% in the EBA 2010 data.

**Recommendations.** (i) Fund the marginal unit of bank liabilities through a Eurosystem refinancing facility at a policy rate, with sovereign collateral and a penalty schedule (option D3). It nests Bocola's own open-economy closure, puts the baseline and the TPI on one funding structure, and makes the deposit rate move only when collateral binds. (ii) Anchor cross-border holdings on *shares* with a return semi-elasticity calibrated to an observable (option P1). Move to fixed shares (P3, two fewer states) once the funding side carries the external margin, if the stress test confirms that the portfolio channel is then small. (iii) Finish the stress test of Section 5: the script is provided, the quick-grid union sweep is done, and the national sweep and the full-grid confirmation remain.

# 1 The two specifications, and what is wrong with them

Notation follows the paper draft. Countries $X \in \{D, F\}$; $\varkappa = \mathcal{M}_F/\mathcal{M}_D = 8$ is the size of $F$ relative to $D$; $p_t$ is the price of the $F$ good in $D$ goods. $R_{X,t} \equiv 1 + r_{X,t}$ is the deposit rate set at $t$ and paid at $t+1$. $\Omega_{X,t,t+1}$ is the banker's stochastic discount factor (Bocola's kernel). $\lambda_X$ is the divertible fraction of assets $\mathcal{A}_{X,t}$, $n_{X,t}$ is net worth, and $\mu_{X,t}$ is the multiplier on the incentive constraint $\alpha_{X,t} n_{X,t} \geq \lambda_X \mathcal{A}_{X,t}$, with Bocola's closed form

$$\mu_{X,t} = \max\Big\{1 - \frac{\mathbb{E}_t[\Omega_{X,t,t+1}] R_{X,t}\, n_{X,t}}{\lambda_X \mathcal{A}_{X,t}},\, 0\Big\}, \qquad r^{wc}_{X,t} = r_{X,t} + \frac{\lambda_X \mu_{X,t}}{\mathbb{E}_t[\Omega_{X,t,t+1}]}. \tag{1}$$

The loan rate $r^{wc}$ (the rate on working-capital loans) is the only channel from the financial block to output, through labour demand $w_{X,t}/(1 + \zeta r^{wc}_{X,t})$, where $\zeta$ is the working-capital share.

## 1.1 Deposit funding

**National market** (the default on the unmerged branch `domestic-funding-fix`): $\text{dep}_{X,t} = P^c_{X,t} HA_{X,t}$ for each $X$, where $HA$ is household deposits in basket units and $P^c$ the consumption price index; $R_{X,t}$ is whatever clears it.

**Union market** (`main`, and the TPI's economy on every branch): one union-wide clearing condition plus real interest parity with the debt-elastic premium of Schmitt-Grohé and Uribe (2003),

$$\text{dep}_{D,t} + \varkappa p_t \,\text{dep}_{F,t} = P^c_{D,t} HA_{D,t} + \varkappa p_t P^c_{F,t} HA_{F,t}, \tag{2}$$

$$R_{D,t} = R_{F,t} \frac{\mathbb{E}_t[p_{t+1}]}{p_t} - \kappa_{\text{nfa}} \frac{V^{\text{flow}}_{D,t}}{\bar{Y}_D}, \qquad V_{t+1} = R_{D,t} V^{\text{flow}}_{D,t}, \tag{3}$$

where $V^{\text{flow}}_{D,t}$ is $D$'s net cross-border deposit position and $\kappa_{\text{nfa}} = 0.01$ (`point_map.py:451`, `calibration/global_projection.py:46`).

**Fault 1: the funding rate is an automatic stabiliser with the wrong sign.** In both closures the deposit rate is the price of a riskless claim whose demand by banks falls when sovereign risk rises (banks shrink and the bond losses hit net worth). The rate therefore falls, and because $r^{wc} = r + \text{spread}$, the fall offsets the spread one for one:

| Headline shock ($p^d = 2\%$/qtr), impact | union, `main` | union, current | national, current |
|---|---|---|---|
| credit spread $\lambda\mu/\mathbb{E}[\Omega]$ (bp/yr) | +79.4 | +76.6 | +65.0 |
| deposit rate $r_D$ (bp/yr) | −47.9 | −46.6 | −33.1 |
| net loan rate $r^{wc}_D$ (bp/yr) | +34.5 | +32.9 | +34.1 |
| share of the spread cancelled | 60% | 61% | 51% |
| output, credit-spread channel (%) | −0.179 | −0.173 | −0.146 |
| output, deposit-rate channel (%) | +0.104 | +0.101 | +0.072 |
| output, total impact (%) | −0.073 | −0.061 | −0.068 |

Full grid (`S_REFINE`=5). "Current" is `domestic-funding-fix` with the household-budget and cross-border $p'/p$ fixes. Source: `results/GLOBAL`, `results/UNION_current`, `results/NATIONAL` run logs, `output_decomposition`.

The credit-spread channel alone ($-0.15$ to $-0.18\%$) lands on Bocola's like-for-like target for his open economy, $-0.157\%$, whose transmission (GHH preferences plus working capital) this model shares. In that economy the funding rate is a world rate, $R = 1/\beta + 0.01\, B^{\text{for}}/\text{gdp}$, which hardly moves on impact. The gap between our impact response and his is therefore, to first order, the endogenous funding rate. (This is an accounting decomposition, not a counterfactual: a closure that pins the rate also changes the general equilibrium. The stress test can measure that directly.)

The fall is also counterfactual. In 2010–12 periphery bank funding costs *rose*: deposit rates in Greece rose relative to the euro area, private deposits ran off, and the Eurosystem (MROs, LTROs, ELA) became the marginal funder. Bank and sovereign credit risk co-moved (Acharya et al., 2014), and sovereign stress reached bank lending through banks' sovereign exposures (Altavilla et al., 2017). The model cannot produce this, because deposits are riskless in both regimes (the household-side "$\pi$-blindness" listed in `CLAUDE.md`) and nothing else connects the deposit rate to the sovereign.

**Fault 2: each closure buys solvability with a device.** *National:* the D market has no external margin except the banks' cross-border sovereign books. Solving it at the 55% haircut needs no-short-sale Fischer–Burmeister pairs on all three bank books, a collapsed TPI band, and a least-squares fallback in the Newton. A purchase barely moves the D bond price, so the TPI cap cannot be held below about 490 bp, and the backstop is evaluated on a different economy (union) from the paper's baseline (national). *Union:* the external position $V$ is a twelfth state whose only anchor is $\kappa_{\text{nfa}}$, a stationarity device borrowed from the small-open-economy literature and never calibrated here. Parity also puts an expectation $\mathbb{E}_t[p_{t+1}]$ inside a residual.

**Fault 3: the closure, not the friction, decides the medium run.** On the current code, cumulative output over 25 quarters is $-3.89$ (national) against $+0.51$ (union); cumulative investment is $-6.00$ against $+4.07$. The impact responses are almost the same. A reader cannot tell from the paper which of these is the model's answer.

## 1.2 Cross-border holdings

The F bank's first-order condition for the D bond, with its non-pecuniary cost, is (`point_map.py:356--357, 465`; the $p_t/p_{t+1}$ factor is in the draft and on `domestic-funding-fix`, *not on* `main`)

$$\mathbb{E}_t\Big[\Omega_{F,t,t+1}\Xi^D_{t+1}\frac{p_t}{p_{t+1}}\Big] = q^D_t\big(\mathbb{E}_t[\Omega_{F,t,t+1}]R_{F,t} + \lambda_F\mu_{F,t}\big)\Big[1 + \psi_F\frac{b^D_{F,t+1} - \bar{b}^D_F}{\bar{B}_D}\Big], \tag{4}$$

with $\psi_F = \psi_D = 2$ and $\bar{b}^D_F = \bar{b}^F_D = 0.196$ (per $D$ capita). Here $\Xi^D$ is the bond's payoff (coupon plus continuation value, times recovery in default), $q^D$ its price, $b^D_F$ the F bank's holding, and $\bar{B}_D$ the SS debt stock.

**What $\psi$ is.** Define the F bank's expected excess return on the D bond, net of its marginal funding cost including the shadow cost of its constraint,

$$e^D_{F,t} \equiv \frac{\mathbb{E}_t[\Omega_{F,t,t+1}\Xi^D_{t+1}p_t/p_{t+1}]}{q^D_t(\mathbb{E}_t[\Omega_{F,t,t+1}]R_{F,t} + \lambda_F\mu_{F,t})} - 1. \tag{5}$$

Then (4) is *exactly*

$$\frac{b^D_{F,t+1} - \bar{b}^D_F}{\bar{B}_D} = \frac{e^D_{F,t}}{\psi_F}. \tag{6}$$

So $1/\psi$ is the semi-elasticity of cross-border holdings, as a fraction of the stock, with respect to the quarterly excess return. At $\psi = 2$, an excess return of 100 bp/yr moves holdings by 0.125% of the stock. Because the D bank prices the bond with no friction, $e^D_F$ is the gap between the two banks' risk-adjusted valuations. That gap is driven by how differently $\Omega_D$ and $\Omega_F$ load on the default state, and by $\mu_D$ against $\mu_F$.

**Four jobs for one parameter.** (a) *Selection.* A single-$\lambda$ banker's value is linear in its portfolio, so at a symmetric SS the split of each bond between the two banks is indeterminate, and off the SS it jumps to a corner. $\psi > 0$ is what makes it determinate. (b) *Elasticity.* By (6), $\psi$ alone scales the retrenchment or flight-home response. (c) *External margin.* Under national funding, $D$'s external position is the net cross-border sovereign book $\text{nfa}_D = p\,q^F b^F_D - q^D b^D_F$. Every unit of the current account therefore passes through (4) and its twin, and $\psi$ becomes the inverse elasticity of $D$'s access to foreign saving. That is the mechanism behind the national "$S = I$ trap" (Fault 3), and it is why one should expect $\psi$ to matter far more under national than under union funding. (d) *Solvability.* In the default branch, at the 55% haircut, F banks want to sell their entire D book. A finite $\psi$ is what keeps that branch interior: the sweep finds no solution at $\psi \leq 0.5$ (Section 5), the same corner that forced the no-short-sale pairs under national funding.

**Calibration faults.** (e) *The anchor is a level.* When the risk shock raises the debt stock, (4) makes F banks resist holding more than $\bar{b}^D_F$, so new issuance is absorbed at home and the F share falls mechanically. That is home bias produced by the units, not by a mechanism. The quick-grid sweep below shows it: a larger $\psi$ gives a *larger* fall in the F share. (f) *The SS position contradicts the data on the D side.* The symmetric gross positions required for a zero net position at $p = 1$ give:

| | model SS | EBA 2011 stress test (Dec 2010) |
|---|---|---|
| F-bank share of bank-held D debt | 20.0% | 12.7% |
| foreign share of the D bank's sovereign book | 20.0% | 0.75% |
| foreign share of the F bank's sovereign book | 2.5% | 2.45% |

EBA: Greek banks hold EUR 54.4bn of Greek and 0.41bn of German sovereigns; German banks hold 7.93bn of Greek and 315.3bn of German (`data/eba_moments.json`, gross direct long).

So the D bank carries a buffer of foreign safe bonds about 27 times the one Greek banks actually had. Under national funding that buffer is $D$'s only way to finance a current-account gap.

## 2 Literature

**Funding rates in intermediary models.** In Gertler and Karadi (2011) and Gertler and Kiyotaki (2010) deposits are riskless and the rate comes from the household Euler equation. Gertler and Kiyotaki (2010) add an interbank market in which a fraction $\omega$ of interbank borrowing escapes the divertibility problem, which spans segmented and frictionless funding. Bocola (2016) keeps

deposits riskless; his open-economy section (§V.C) closes the model with a world rate plus a debt-elastic premium in the manner of Schmitt-Grohé and Uribe (2003). In emerging-market models with banks, Aoki et al. (2016) let banks borrow both domestically and abroad, with foreign funding more easily diverted; the convex divertibility $\theta(1 + \frac{\gamma}{2}x^2)$, where $x$ is the foreign share, prices the funding mix. Akinci and Queralto (2018) and Dedola et al. (2013) build two-country intermediary models with cross-border funding and holdings; Kollmann et al. (2011) model a global bank. Country-spread closures, in which the borrowing rate is the world rate times a country premium, are standard since Neumeyer and Perri (2005) and Uribe and Yue (2006). In Neumeyer and Perri (2005) the spread works through working capital, which is this model's own output channel. On the sovereign-to-private pass-through, Corsetti et al. (2013) posit that private funding costs load a fraction of the sovereign spread. Deposit-supply frictions and convenience yields (Drechsler et al., 2017; Begenau, 2020) make deposit rates sticky relative to the safe rate. On the euro-area plumbing, Drechsler et al. (2016) show that weakly capitalised banks borrowed most from the lender of last resort against risky collateral, and Crosignani et al. (2020) show that LTRO funding was used to buy domestic sovereigns. Gertler et al. (2016) split bank funding into retail and wholesale and add runs.

**Cross-border portfolios.** Devereux and Sutherland (2011) and Tille and van Wincoop (2010) determine portfolios from second- and third-order terms around a deterministic SS. That is not needed in a global solution, but they show why first-order models need an extra device. Portfolio-balance and limited-risk-bearing closures, from Blanchard et al. (2005) to Gabaix and Maggiori (2015) and Itskhoki and Mukhin (2021), make holdings a finite-elastic function of expected excess returns, with the elasticity tied to intermediaries' risk capacity. Coeurdacier and Rey (2013) survey home bias. Demand-system estimation (Koijen and Yogo, 2019; Koijen et al., 2021) measures exactly the object in (6), the elasticity of a sector's holdings to yields, for euro-area sovereigns by holder sector. For the euro crisis, Broner et al. (2014) and Brutti and Sauré (2016) document the repatriation of periphery debt to domestic holders. Explanations include creditor discrimination (Broner et al., 2014), risk-shifting or carry trades by weak banks (Acharya and Steffen, 2015), moral suasion (Ongena et al., 2019), financial repression (Becker and Ivashina, 2018), and the bank–sovereign doom loop (Farhi and Tirole, 2018; Gennaioli et al., 2014). Altavilla et al. (2017) and Battistini et al. (2014) measure the exposures and their transmission. Value-at-risk constraints (Adrian and Shin, 2014) give a structural reason for risk-sensitive exposure limits.

## 3 Alternatives for the deposit funding rate

Each option is stated with what it does to the period map (`point_map.py`): its residuals, unknowns and states.

### D1. Two-tier funding with costly cross-border wholesale funding

The D bank funds itself with net worth, national retail deposits $\text{dep}_D$ at $R_D$, and union wholesale funding $w_D$ (D goods) at $R^w$. Let $x_t \equiv w_{D,t}/\mathcal{A}_{D,t}$. In the spirit of Aoki et al. (2016), wholesale funding raises divertibility:

$$\alpha_t n_t \;\geq\; \lambda\,\Theta(x_t)\mathcal{A}_t, \qquad \Theta(x) = 1 + \tfrac{\gamma}{2}x^2. \tag{7}$$

Because $\Theta(x)\mathcal{A}$ is homogeneous of degree one in $(\mathcal{A}, w)$, Bocola's closed form survives with $\lambda \to \lambda\Theta(x)$:

$$\mu_t = \max\Big\{1 - \frac{\mathbb{E}_t[\Omega] R_{D,t}\, n_t}{\lambda\Theta(x_t)\mathcal{A}_t}, 0\Big\}, \qquad \alpha_t = \frac{\mathbb{E}_t[\Omega] R_{D,t}}{1-\mu_t}, \tag{8}$$

$$\mathbb{E}_t[\Omega(R^K_{t+1} - R_{D,t})] = \lambda\mu_t\big(1 - \tfrac{\gamma}{2}x_t^2\big) \quad \text{(and likewise for bonds and loans)}, \tag{9}$$

$$R_{D,t} - R^w_t = \frac{\lambda\gamma\,\mu_t\, x_t}{\mathbb{E}_t[\Omega]} + \kappa_w(x_t - \bar{x}), \tag{10}$$

with retail clearing $\text{dep}_{D,t} = P^c_{D,t} H A_{D,t}$, parity for the union lender $R^w_t = R_{F,t}\mathbb{E}_t[p_{t+1}]/p_t$, and $F$'s clearing net of its households' wholesale lending. The small $\kappa_w$ term is an SGU device that matters only where $\mu = 0$. *Nesting:* $\gamma \to \infty$ gives $x \to 0$ (national); $\gamma = 0$ gives $R_D = R^w$ (union). *Sign:* when $\mu_D$ rises, (10) pushes the domestic deposit rate *up* relative to the union rate, the deposit competition of 2010–12. *Calibration in closed form:* at the SS, dividing (10) by the credit spread gives $\gamma = (R_D - R^w)/[\,\bar{x}\,(r^{wc} - r_D)\,]$ up to the $x^2$ term. That uses three observables: the retail deposit spread over Euribor (ECB MIR statistics), the lending spread (already targeted), and the non-resident wholesale share of Greek bank funding (Bank of Greece MFI balance sheets). *Cost:* one unknown and one residual more (15); $V$ is reused as the wholesale position.

## D2. Defaultable deposits: sovereign risk in the deposit rate

Deposits pay a fraction $h^{\text{dep}} \in (0, 1]$ of their promised return in the default regime (bail-in, as in Cyprus in 2013, or redenomination; Krishnamurthy et al., 2018 measure the latter). The household Euler weights the default branch:

$$x^{-\sigma}_{D,t} = \beta\,\mathbb{E}_t\Big[\big(1 - (1-h^{\text{dep}})d_{t+1}\big) R_{D,t} \frac{P^c_{D,t}}{P^c_{D,t+1}}\, x^{-\sigma}_{D,t+1}\Big], \tag{11}$$

where $x$ is the GHH composite and $d_{t+1}$ the default indicator. The bank's deposit leg is written down by the same factor in that branch, $\mathfrak{P}^{d'=1}_{D,t+1} = h^{\text{dep}} R_{D,t}\text{dep}_{D,t} - R^{wc}_{D,t} L_{D,t}$, so the depositor's loss is the bank's gain and nothing leaks from the budget. To first order,

$$r_{D,t} - r^{h=1}_{D,t} \;\approx\; \pi_t(1-h^{\text{dep}})\,\frac{\mathbb{E}_t[x^{-\sigma}_{t+1} \mid d'=1]}{\mathbb{E}_t[x^{-\sigma}_{t+1}]}, \tag{12}$$

so the deposit spread is the default probability times the deposit loss-given-default, weighted by marginal utility. Matching the reduced form of Corsetti et al. (2013), deposit spread $= \xi\times$ sovereign spread, gives $h^{\text{dep}} \approx 1 - \xi(1-0.45)$. *Cost:* no new state and no new unknown. The default branch already exists in the quadrature; only the next-period state for $d' = 1$ becomes regime-specific inside `_continuation`, which builds the state per regime anyway. *Caveat:* the bail-in transfer cushions $n_D$ in default, so it lowers $\Omega^d$ and the bank-side risk premium; redenomination without a bank-side gain is the other polar case. Works with any deposit market.

## D3. Eurosystem refinancing at a policy rate, against sovereign collateral

The D bank's marginal funding is a Eurosystem facility $\ell_t$ (D goods). The facility is priced on a penalty schedule (ELA lends above the main refinancing rate) and limited by haircut-adjusted

collateral:

$$\mathcal{A}_{D,t} = n_{D,t} + \mathrm{dep}_{D,t} + \ell_t, \qquad R^{\ell}_t = \bar{R} + \kappa_\ell \frac{\ell_t - \bar{\ell}}{\bar{Y}_D}, \qquad \ell_t \le (1 - h^c)\, q^D_t b^D_{D,t+1}, \tag{13}$$

$$R_{D,t} = R^{\ell}_t + \frac{\eta_t}{\mathbb{E}_t[\Omega_D]}, \qquad \eta_t \ge 0, \quad (1-h^c) q^D_t b^D_{D,t+1} - \ell_t \ge 0, \quad \eta_t \left[(1-h^c) q^D_t b^D_{D,t+1} - \ell_t\right] = 0, \tag{14}$$

$$q^D_t \left(\mathbb{E}_t[\Omega_D] R_{D,t} + \lambda \mu_{D,t} - (1-h^c)\eta_t\right) = \mathbb{E}_t[\Omega_D \Xi^D_{t+1}]. \tag{15}$$

Here $\eta$ is the collateral multiplier (in the bank's value units), $h^c$ the collateral haircut, and $\kappa_\ell$ the slope of the penalty schedule. The Eurosystem funds $\ell$ with reserves held by $F$ banks at $R_F$ and remits the facility's P&L, including the terms-of-trade valuation, by the capital key, through the same accounts the TPI already uses ($\Pi^{cb}$, `tpi_key_D`). That is what makes a rate fixed in D goods consistent in a two-country model: the valuation profit is assigned.

*What it does.* While collateral is slack, $R_D = R^{\ell}$ is set by policy: the deposit rate stops absorbing the shock, households are no longer forced to hold whatever deposits the bank wants, and $D$ can draw on union saving through the facility (TARGET2 in all but name). When the sovereign is marked down, collateral binds and $\eta > 0$, which raises $R_D$: the pass-through has the right sign and runs through $q^D$. By (15), binding collateral also raises the D bank's valuation of its own sovereign, so repatriation becomes an outcome rather than a by-product of units (Crosignani et al., 2020; Drechsler et al., 2016). *Nesting:* with collateral slack and $\bar{\ell} = 0$, $R^{\ell} = \bar{R} + \kappa_\ell \ell / \bar{Y}_D$ is Bocola's open-economy closure ($\kappa_\ell = 0.01$, with $\ell$ in the role of his foreign debt); $\kappa_\ell \to \infty$ with $\bar{\ell} = 0$ is the national market. *Calibration:* $h^c$ comes from the published Eurosystem haircut schedule for the rating class; $\bar{\ell}$ from Greek banks' Eurosystem borrowing before the crisis; $\kappa_\ell$ from the ELA–MRO spread. *Cost:* one state (the facility obligation, replacing the inert $V$ under national funding), one unknown and one Fischer–Burmeister pair, a device the code already uses. The national solver devices, the collapsed TPI band and the separate union baseline should no longer be needed, because the external margin now exists in every economy.

| | sign of $\Delta r_D$ on risk | nests | new parameters (data) | states / unknowns |
|---|---|---|---|---|
| national (now) | falls | — | — | 12 / 14, 3 devices |
| union (now) | falls | — | $\kappa_{\mathrm{nfa}}$ (none) | 12 / 14 |
| D1 wholesale | rises if $\mu \uparrow$ | national, union | $\gamma, \bar{x}$ (MIR, BoG) | 12 / 15 |
| D2 defaultable | rises with $\pi$ | any closure | $h^{\mathrm{dep}}$ (MIR spread) | unchanged |
| D3 Eurosystem | rises if collateral binds | national, Bocola §V.C | $h^c, \bar{\ell}, \kappa_\ell$ (ECB) | 12 / 15, 1 FB pair |

**Recommendation: D3.** It is the only option in which the marginal funding rate is a policy rate rather than an equilibrium price, which removes Fault 1 by construction outside the collateral-binding region. It removes Fault 2, because baseline and TPI share one funding structure with the central bank in both. Its three parameters have published counterparts. D2 can be layered on top if the paper wants deposit flight. D1 is the cleanest dial for a robustness section, since $\gamma$ moves the economy continuously from national to union.

# 4 Alternatives for cross-border holdings

### P1. Share anchor with an identified return semi-elasticity

Replace the level anchor in (4) by the holder's share of the outstanding stock, $s_{F,t} \equiv b^D_{F,t+1}/B^D_{t+1}$:

$$\mathbb{E}_t\Big[\Omega_F \Xi^D_{t+1} \frac{p_t}{p_{t+1}}\Big] = q^D_t \big(\mathbb{E}_t[\Omega_F] R_{F,t} + \lambda \mu_{F,t}\big)\big[1 + \tilde{\psi}_F (s_{F,t} - \bar{s}_F)\big] \iff s_{F,t} - \bar{s}_F = \frac{e^D_{F,t}}{\tilde{\psi}_F}. \tag{16}$$

New issuance is then shared at $\bar{s}_F$ unless returns say otherwise. $1/\tilde{\psi}$ is the share's semi-elasticity, which can be read from demand-system estimates for foreign banks (Koijen et al., 2021), or from the 2010–11 change in German banks' Greek exposure (EBA 2011 stress test against the EBA 2011 capital exercise; BIS consolidated banking statistics) divided by the model-implied $\Delta e$ at the observed spread. Set $\bar{s}_F = 12.7\%$ (EBA). Note that matching the D side (0.75%) breaks the zero net position at $p = 1$. Under D3 the facility can carry $D$'s SS net debt, so the asymmetry is absorbed on the funding side without moving $p_{\text{ss}}$. *Cost:* none; same residuals and states.

### P2. Limited risk-bearing capacity (a risk budget)

GK kernels price $D$'s default as nearly idiosyncratic for an $F$ banker, so the model has no force that makes foreign banks *shed* when risk rises. A risk budget in the spirit of Adrian and Shin (2014) and Gabaix and Maggiori (2015) adds one. Let $\chi_{F,t} \equiv q^D_t b^D_{F,t+1}/(\varkappa p_t n_{F,t})$ be the exposure as a share of $F$ net worth, and let $\mathcal{V}_t \equiv \text{Var}_t\big(\Xi^D_{t+1} p_t/(q^D_t p_{t+1})\big)$ be its conditional return variance, computed on the existing $s' \times d'$ quadrature at no extra cost. Then

$$\chi_{F,t}\,(\bar{\sigma}^2 + \mathcal{V}_t) = \chi_{\text{ss}}\,\bar{\sigma}^2 + \frac{e^D_{F,t}}{\gamma_F}. \tag{17}$$

At the deterministic SS ($\mathcal{V} = 0$, $e = 0$) $\chi = \chi_{\text{ss}}$, so it is determinate. When $\pi$ rises, $\mathcal{V}_t$ rises roughly in proportion to $\pi_t(1 - 0.45)^2$ and the exposure falls by the factor $\bar{\sigma}^2/(\bar{\sigma}^2 + \mathcal{V}_t)$; when $F$ net worth falls, the level falls too, which carries contagion. $\bar{\sigma}^2$ is the variance at which the budget halves the exposure; $\gamma_F$ is the return sensitivity. Both are identified by the 2010–11 retrenchment together with its return. *Cost:* same residual slots, a few quadrature sums.

### P3. Fixed shares (segmentation, the $\psi \to \infty$ limit)

$b^D_{F,t+1} = \bar{s}_F B^D_{t+1}$ and $b^F_{D,t+1} = \bar{s}_D B^F$ ($B^F$ is fixed in the code). Residuals 9 and 13 and their two unknowns go, and the states $b_{DF}$ and $b_{FD}$ become functions of $b_{DD}$ and $M^{cb}$, so the grid goes from 12 states to 10. On the measured scaling (the solve is $\propto n^2$, with 115 points falling to 95), the full grid is about 30% cheaper. F banks still take marked-to-market losses on $\bar{s}_F$ of the D stock, so contagion survives; what is lost is cross-bank rebalancing, and with it the split of the TPI's purchases by seller. *Only admissible with an external funding margin* (union, D1 or D3). Under national funding it closes the economy to foreign saving altogether.

**Recommendation: P1 now; P3 once D3 is in, if Section 5 finds the portfolio channel small there; P2 if retrenchment dynamics become a target.** The general point is that the model needs exactly one external adjustment margin with a calibratable elasticity. Today, depending on the deposit closure, that margin is either a frictionless deposit position (union, $\kappa_{\text{nfa}}$)

or the $\psi$-governed bond books (national). D3 puts it on the funding side, where 2010–12 actually put it, and so leaves $\psi$ with one job, not three.

## 5 Stress test: how much does $\psi$ drive the dynamics?

**Design.** The question is quantitative and global, so each point is a full re-solve, never a perturbation of fixed rules.

1. *Sweep.* $\psi \in \{0.25, 0.5, 1, 2, 4, 8, 32\}$ on a log grid, with both $\psi_F$ and $\psi_D$ moved together (then one at a time). Solve the calibrated point cold, then walk outward, warm-starting each solve from its neighbour: a continuation in $\psi$ that keeps every Newton inside its basin. Quick grid ($\mu = 1$, 25 points) for the sweep; the full grid (`S_REFINE`=5) at $\{0.5, 2, 8\}$ to confirm that the elasticities are grid-independent.

2. *Read.* Bocola's differenced IRF from each solution's own stochastic rest point, with the headline shock $p^d = 2\%$/qtr for 25 quarters. Re-locate the rest point for every $\psi$: it moves with $\psi$, and differencing against a stale one charges the drift to the shock.

3. *Metrics.* Impact, trough and cumulative output; cumulative investment; impact $n_D$, $n_F$ and $q^D$; the credit spread and deposit rate on impact; the F-bank share of $D$ debt at $q4$; $D$'s net external position at $q8$; and the model's retrenchment elasticity (pp of $D$ debt shed by F banks per 100 bp of sovereign spread), which is the moment to set against the data.

4. *Sensitivity.* The log-elasticity at the calibrated point, $\varepsilon_m = \partial \ln |m| / \partial \ln \psi$, from the two neighbours, and the sweep range $(\max m - \min m)/|m(\psi_{\text{ref}})|$. Use elasticities rather than raw changes so that dials measured in different units can be ranked against each other (step 7).

5. *Isolation.* Two decompositions. (a) *Portfolio channel*: $m(\psi) - m(\psi \to \infty)$, the IRF relative to fixed holdings (P3 is that limit). (b) *Within-period versus expectations*: re-clear the period map exactly at impact under each $\psi$ while holding the reference continuation rules fixed. The ratio of that change to the fully re-solved change is the share of $\psi$'s effect that runs through the current FOC; the remainder runs through expectations of future rebalancing.

6. *Accuracy guards.* Each point records the solver verdict and worst residual (Bocola's $\sum F^2 \leq m \cdot 10^{-18}$), the largest box clip along the IRF (the bond-state band is $\pm 12\%$, so a small $\psi$ can push holdings out of the box), and the impact read two ways, fitted and exactly cleared. A $\psi$ effect smaller than that gap is not identified at the current resolution. On the full grid, add the Euler-error report on the ergodic set.

7. *Ranking against the other dials.* Repeat the sweep for $\kappa_{\text{nfa}}$, $f$, the credit-spread target, $\zeta_{wc}$, the Jermann $\xi$, $\sigma_s$ and the recovery rate, and rank by $|\varepsilon_{\text{cumY}}|$. That gives a tornado table, from one-at-a-time sweeps. For interactions, a Morris elementary-effects screen (Morris, 1991) with ten trajectories over these eight dials is about 90 quick solves, roughly an overnight run at about 1–5 minutes each.

8. *Interaction with the funding closure.* The whole sweep in both deposit markets. Under national funding, $\psi$ is predicted (Section 1) to govern the medium run; under union funding only the composition. The ratio of the two elasticities is the cleanest test of that claim.

**Implementation.** `Claude files/diagnostics/portfolio_stress/psi_sweep.py` does steps 1–6 for any calibration key, and step 7 as a ranking table (`--compare`) across saved sweeps. Results go to `results/STRESS/<key>_<market>_s<refine>.json`, with a six-panel figure. Typical calls:

```
S="Claude files/diagnostics/portfolio_stress/psi_sweep.py"
python3 "$S"                                    # psi, quick grid
python3 "$S" --values 0.5 8 --s-refine 5        # full-grid confirmation
python3 "$S" --param kappa_nfa --values 0.0025 0.04
python3 "$S" --set deposit_market=national      # domestic-funding-fix only
python3 "$S" --compare                          # rank all saved sweeps
```

`--set deposit_market=national` works only on `domestic-funding-fix`; on `main` the key does not exist and the run is the union economy.

## First results: quick grid, union deposits, `main`

| $\psi$ | $Y_0$ fit % | $Y_0$ exact % | $\sum Y$ 25q | $\sum I$ 25q | $n_{D,0}$ % | spread$_0$ bp/yr | $r_{D,0}$ bp/yr | $\Delta s_F$ q4 pp | nfa q8 % $Y$ | $Y_0$ exact $\psi$=2 rules |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | -0.047 | -0.060 | +1.64 | +8.24 | -5.09 | +94.8 | -47.8 | -0.342 | +0.701 | -0.069 |
| **2** | -0.064 | -0.076 | +1.63 | +8.49 | -5.20 | +100.1 | -50.6 | -0.445 | +0.756 | -0.076 |
| 4 | -0.051 | -0.062 | +1.90 | +8.98 | -5.20 | +102.3 | -52.3 | -0.505 | +0.745 | -0.079 |
| 8 | -0.047 | -0.056 | +2.03 | +9.19 | -5.22 | +103.7 | -53.2 | -0.537 | +0.744 | -0.081 |
| 32 | -0.043 | -0.052 | +2.13 | +9.36 | -5.23 | +104.9 | -53.9 | -0.563 | +0.743 | -0.083 |
| range / $\lvert m(2)\rvert$ | 33% | | 30% | 13% | 3% | 10% | 12% | 50% | 7% | |

Headline shock $p^d = 2\%$/qtr, differenced IRF from each solution's own rest point. Coarse grid (25 points), a preview: at $\psi = 2$ the full grid gives $-0.073\%$ against $-0.064\%$ here. The last column re-clears the period map exactly at impact under each $\psi$ with the $\psi = 2$ continuation rules held fixed (the within-period effect alone). The range row is $(\max m - \min m)/|m(\psi{=}2)|$ over the solved values. No solution at $\psi = 0.5, 0.25$: the warm walk (1, 2 and 4 log-steps from $\psi = 1$) and the cold ladder both fail.

| dial (values swept) | ln range | $Y_0$ range | $\sum Y$ range | $\Delta s_F$ range | nfa q8 range |
|---|---|---|---|---|---|
| $\psi$ (1–32) | 3.47 | 33% | 30% | 50% | 7% |
| $\kappa_{\text{nfa}}$ (0.0025–0.04) | 2.77 | 110% | 56% | 3% | 9% |
| per unit of ln(dial): $Y_0$: $\psi$ 9.5%, $\kappa_{\text{nfa}}$ 40%; $\sum Y$: $\psi$ 8.7%, $\kappa_{\text{nfa}}$ 20% | | | | | |

Same grid, shock and reads as above; ranges as a share of the value at the calibrated point. $\kappa_{\text{nfa}}$: impact output $-0.085$, $-0.064$, $-0.014\%$ at 0.0025, 0.01, 0.04; cumulative output +2.02, +1.63, +1.10 (`results/STRESS/kappa_nfa_default_s0.json`).

### Reading.

1. *Solvability bounds $\psi$ from below.* No solution exists at $\psi \leq 0.5$: the warm walk from $\psi = 1$ (in one, two and four log-steps) stalls at $\max|F| \approx 3 \cdot 10^{-2}$, and the cold ladder produces a non-finite iterate in the default-branch haircut homotopy. In that branch the F bank wants to sell its whole D book; with the $10^{-4}$ floors in place of a complementarity pair the Jacobian degenerates. This is the national branch's failure, reached here through the portfolio cost.

The interesting low-$\psi$ region cannot be explored without porting the no-short-sale pairs to the union economy.

2. *Under union funding, $\psi$ governs composition, not the current account.* The F share moves by half its value across the sweep, while $D$'s net external position moves by 7%, because the deposit position $V$ absorbs the external adjustment whatever $\psi$ is (the six-panel figure, `results/STRESS/psi_bF_D_default_s0.png`). This is the prediction of Section 1, job (c), read in reverse.

3. *Output: a third over a 32-fold range, with a bump at the calibrated point.* A stiffer friction gives a smaller impact fall and a larger later boom. Above $\psi = 2$ both move monotonically: the one-sided elasticity between 2 and 32 is $-0.15$ for $|Y_0|$ and $+0.10$ for cumulative output. The $\psi = 1$ point breaks monotonicity on impact. That is *not* multiplicity: cold-solving $\psi = 4$ reproduces the warm solution exactly ($-0.0514\%$), and walking $4 \to 2$ warm reproduces the cold $\psi = 2$ ($-0.0644\%$). On 25 points it may be an interpolation effect at the KKT kink; the full-grid points $\{1, 2, 4\}$ decide.

4. *The mechanism runs through expectations.* Holding the $\psi = 2$ rules fixed and changing $\psi$ only in the period map moves impact output the *other* way ($-0.069\%$ at $\psi = 1$ to $-0.083\%$ at $\psi = 32$): today a stiffer friction stops F banks buying at the dip, so the D bank absorbs more. The fully re-solved effect has the opposite sign, so what $\psi$ does to the IRF is mostly what it does to expected future rebalancing, which a fixed-rule read excludes.

5. *The level anchor shows up as predicted.* The F share falls *more* as $\psi$ rises ($-0.34$ to $-0.56$ pp at q4), the opposite of what a stiffer rebalancing friction should do, because the friction holds the level while the debt stock grows. At $\psi = 1$ the F share rises on impact (F banks buy at the dip) before falling.

6. *The funding closure dominates.* Per log unit, $\kappa_{\text{nfa}}$ moves impact output four times as much as $\psi$, and cumulative output twice as much. A stationarity device borrowed from the small-open-economy literature, never calibrated here, is the most powerful dial of the two. Its effect does not run through the loan rate (impact $r^{wc}$ is +50 bp/yr at $\kappa = 0.0025$ and +48 at 0.04) but through the terms of trade and labour supply, as the external margin tightens. The deposit-rate channel of the decomposition (+0.104 pp) is likewise about five times the whole $\psi$ range of impact output (0.021 pp). For the *union* economy, then, the answer to Issue 2 is that $\psi$ drives the cross-border composition strongly and the output path moderately, and Issue 1 is first-order. Under national funding the prediction is the reverse; that sweep is the next run.

## 6 Next steps

1. Merge or port the cross-border $p'/p$ fix to `main` before quoting any $\psi$ result; the sweep above is on `main`'s FOC.

2. Run the sweep under national funding on `domestic-funding-fix`, and the full-grid confirmation at $\{0.5, 2, 8\}$.

3. Pull the four data moments: the GR–euro area deposit spread (ECB MIR), the non-resident wholesale share and Eurosystem borrowing of Greek banks (Bank of Greece, ECB), German banks' Greek exposure in 2011 (EBA capital exercise, BIS CBS), and the ELA penalty spread.

4. Implement D3 behind a `deposit_market="eurosystem"` switch (the N1–N5 nesting tests extend directly: $\kappa_\ell \to \infty$ must reproduce national), then P1, re-run the sweep, and decide on P3.

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